

Joseph Kerkhof

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SUMMARY

Senior Backend Engineer with 7+ years designing secure, reliable web and cloud systems. Built a production AI workflow platform, architected Kubernetes observability infrastructure, and delivered financial products with 90%+ test coverage and sub-second response times. Deep expertise in PHP/Laravel, system design, distributed architectures, and backend fundamentals, with hands-on experience in Go and Python. Uses Claude Code and OpenCode with MCP integrations, repository context files, specialized skills, and automated guardrails to streamline development and review.

TECHNICAL SKILLS

Backend & Data: PHP, Laravel, Go, Python, PostgreSQL, MySQL/MariaDB, Redis, Snowflake, REST APIs

AI Engineering: Laravel AI, LangGraph, Claude Code, OpenCode, MCP Integrations, Agent Orchestration

Cloud & Delivery: AWS, Cloudflare, Linux, Docker, Kubernetes, GitHub Actions, CI/CD

Architecture & Quality: DDD, CQRS, Microservices, Pest, PHPUnit, PHPStan, Pint, TDD, Load Testing

Observability & Identity: OpenTelemetry, ClickStack/HyperDX, Distributed Tracing, Enterprise SSO

WORK EXPERIENCE

Active Engagement

Remote, USA

Senior Developer

Oct 2025 – Present

Tech Stack: Laravel, Laravel AI, PostgreSQL, Python, Kubernetes, OpenTelemetry

- Started and architected a modular Laravel AI platform from the ground up, establishing reusable workflows and modules now developed by three engineers to automate copywriting and an 11-stage email workflow across five business systems.
- Engineered Anthropic-backed orchestrator and specialist agents with tools, persistent multimodal conversations, structured outputs, token-cost telemetry, and deterministic fallbacks; also built the first version of the email workflow in Python/LangGraph and ran it in production pilots.
- Built and refined GitHub Actions CI/CD with Pint, maximum-level PHPStan/Larastan, TypeScript and security checks, 1,500+ Pest tests across five layers, sharded browser tests, atomic releases, and rollback support.
- Designed and deployed ClickStack/HyperDX on autoscaling Kubernetes with Terraform and Helm, unifying logs, metrics, traces, RUM/session replay, AI cost telemetry, dashboards, alerts, uptime checks, backups, SSO, configuration-as-code, and documented recovery procedures.
- Translated sparse product direction into iterative mockups with Claude Design, defined requirements with stakeholders, and guided junior engineers through PR reviews and daily technical feedback.

Teamworks

Remote, USA

Senior Software Engineer

May 2024 – Sep 2025

Tech Stack: Laravel, PHP, MySQL, Redis, AWS, Kubernetes, CI/CD

- Architected and delivered GM budget management product from the ground up in 3 months, processing athlete compensation across multiple universities with 90% test coverage and daily production deployments via automated CI/CD pipelines.
- Delivered integrated payment systems in 8 months ahead of the NCAA's July 2025 regulatory deadline, using Domain-Driven Design, CQRS, and action-based architecture to keep complex regulatory logic maintainable and testable.
- Engineered a high-performance RESTful API achieving sub-second response times through optimized database queries and asynchronous processing with Redis queues, implementing comprehensive observability via structured logging, distributed tracing, and StatsD metrics.

Prodege

PHP Developer

Remote, USA

Mar 2023 – May 2024

Tech Stack: PHP, Laravel, MySQL, Google Ads SDK, Cron Jobs

- Built an automated Google Ads campaign management system that cut launch time from 8 hours to 30 minutes and prevented \$20K–\$30K in monthly wasted spend through real-time budget optimization and alerting.
- Modernized legacy PHP cron job infrastructure to Laravel 10, eliminating 15+ monthly job failures and reducing system maintenance time through improved error handling and monitoring.
- Optimized MySQL queries on 200GB+ tables, reducing query execution time from 45 seconds to under 3 seconds and improving dashboard load times by 85% through table redesign and proper indexing strategies.
- Identified and remediated critical vulnerabilities including remote code execution, SQL injection, and exposed credentials.

Clockwork

Senior Software Engineer

Minneapolis, MN

Mar 2019 – Mar 2023

Tech Stack: PHP, MySQL, AWS Serverless, React, TypeScript, Jest, Load Testing Tools

- Built web products serving 50K+ users as team Delivery Lead, managing sprint planning for 6-person team while architecting React/TypeScript frontends.
- Load-tested high-traffic website processing 100K+ daily requests to identify application and infrastructure bottlenecks, implementing horizontal scaling solutions that improved response times by 60% and eliminated timeout errors.
- Maintained a PHP/MySQL promotion engine serving 350K+ accounts and thousands of daily transactions with 99.9% uptime; participated in on-call support for Minnesota Lottery applications.

EDUCATION

University of Wisconsin — Oshkosh

Oshkosh, WI

M.Sc. in Data Science (Part-time); GPA: 4.00/4.00

Sep 2017 – Jun 2021

B.Sc. in Computer Science

Sep 2011 – Jun 2016

PROJECTS

Omnibus Config File Validator | [GitLab](#)

Tech Stack: Golang, Cobra

- Built a Go CLI that validates GitLab Omnibus configurations for security issues and common misconfigurations, with automated testing, linting, and builds in GitLab CI.

GitHub RSS | [GitHub](#)

Tech Stack: PHP, Laravel, Inertia, Vue.js

- Built a GitHub activity feed service around interface-driven dependencies and in-memory test doubles, enabling fast, isolated unit tests while supporting REST APIs with limit-offset and cursor pagination.